PROJECT PRESENTATION

DeepOCR

Deep Learning-Based Intelligent OCR System for Handwritten and Complex Document Recognition

5 recognition models

Consensus voting

Review studio



October 2026

Why OCR still fails on real documents

Clean printed pages are easy. Real documents are not.

Handwriting

Every writer is different; letters touch and slant.

Poor scans

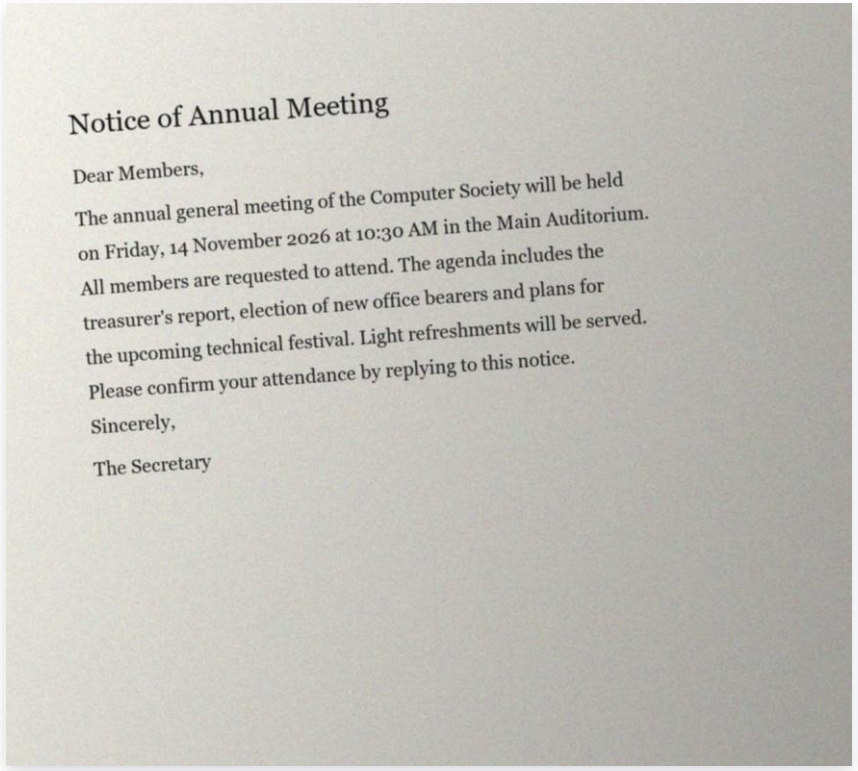
Noise, blur, shadows, low contrast and faint ink.

Skew and layout

Tilted pages, tables, forms, checkboxes, columns.

No sense of doubt

Tools never say which words may be wrong.



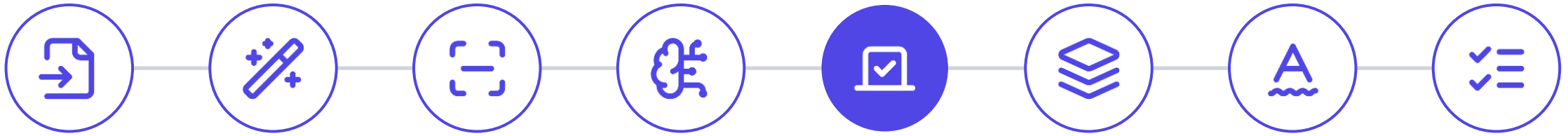
A test input: 4° skew, noise and uneven lighting

Objectives from the project proposal

1. Extract text from document images
2. Recognise handwritten and printed text
3. Clean noisy, blurred and skewed images
4. Detect text regions in complex documents
5. Analyse the document layout
6. Compare deep-learning recognition models
7. Evaluate with standard metrics (CER, WER)
8. Build a user-friendly interface

System overview: an eight-stage pipeline

From an uploaded file to reviewed, structured text



Input
PDF or image, up to 100 pages

Clean up
Deskew, denoise, CLAHE contrast

Detect lines
DBNet finds every text line

Recognise
CNN-BiLSTM, CRNN or TrOCR

Vote
Five models vote on each word

Layout
Headings, tables, checkboxes

Correct
Fix 0/O, 1/l and spelling

Review
Overlays, review queue, export

Only the Consensus engine runs the vote; single models go straight from recognition to layout.

Built with Python · PyTorch · OpenCV · Hugging Face Transformers · PaddleOCR · FastAPI · React + TypeScript

Data: real handwriting plus synthetic print

10,373

IAM handwriting lines · 657 writers



Train 6,482 · Val 976 · Test 2,915

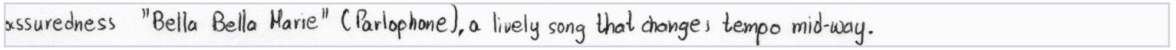
6,500

Synthetic printed and script lines · 32 fonts



Train 5,005 · Val 520 · Test 975

IAM handwriting: “assuredness " Bella Bella Marie " (Parlophone), a lively song that changes tempo mid-way.”



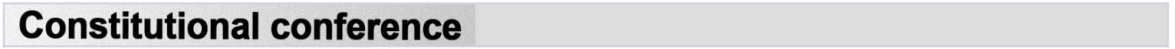
IAM handwriting: “CHRIS CHARLES, 39, who lives in Stockton-on-Tees, is an accountant.”



IAM handwriting: “Become a success with a disc and hey presto! You're a star.... Rolly sings with”



Synthetic: “Constitutional conference”



Synthetic: “in June, made it”



Synthetic: “+91 94065 82140”

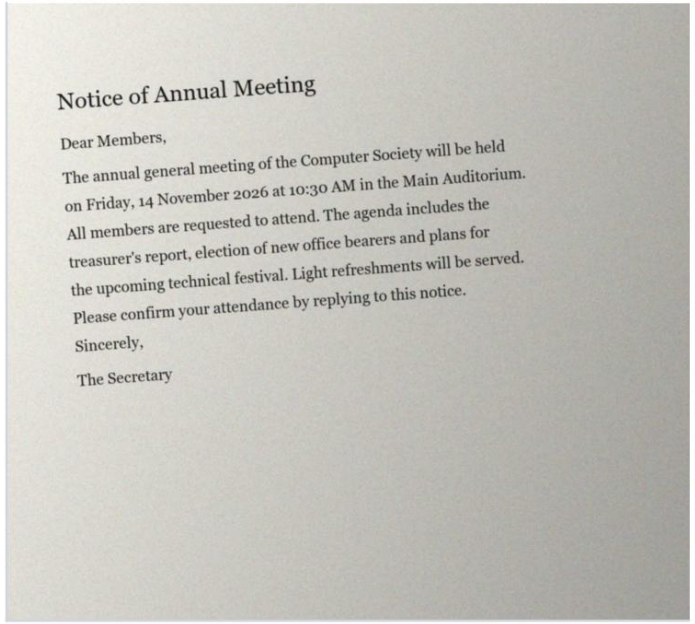


Writer-independent split: test writers never appear in training.

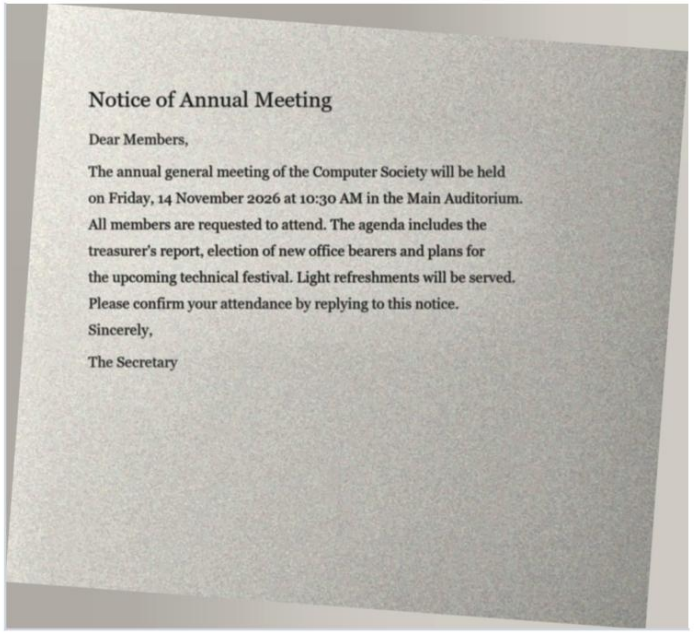
Augmentation: rotation, scaling, noise, blur, brightness, perspective.

Cleaning the page before reading it

Original: skewed, noisy, uneven lighting



After deskew (-4.2°), denoise and CLAHE



65.5% → 18.9%

CER of the CNN-BiLSTM on lines tilted by 8°, without and with our deskew step

Deskew

Projection profile finds the tilt (±12°) and straightens the page

Denoise

Gaussian and median filtering remove speckle

Contrast

CLAHE lifts faint ink without boosting noise

Binarise

Adaptive threshold evens out phone-photo lighting

Three models, from baseline to Transformer

BASELINE

CNN-BiLSTM

4 conv blocks + 2-layer BiLSTM + CTC

Parameters
2.39 M

Input
32 × 512 grey

Training
From scratch, 124 min on CPU

Speed
12 ms per line

MODEL 2

CRNN

7-layer VGG CNN + BiLSTM + CTC (Shi et al.)

Parameters
2.78 M

Input
32 × 512 grey

Training
From scratch, 130 min on CPU

Speed
16 ms per line

ADVANCED

TrOCR (fine-tuned)

Vision Transformer encoder + text decoder

Parameters
61.6 M

Input
384 × 384 RGB

Training
Pretrained, fine-tuned 44 min

Speed
415 ms per line

Also compared: PaddleOCR (industrial PP-OCR) as a reference, and the Consensus vote of all five

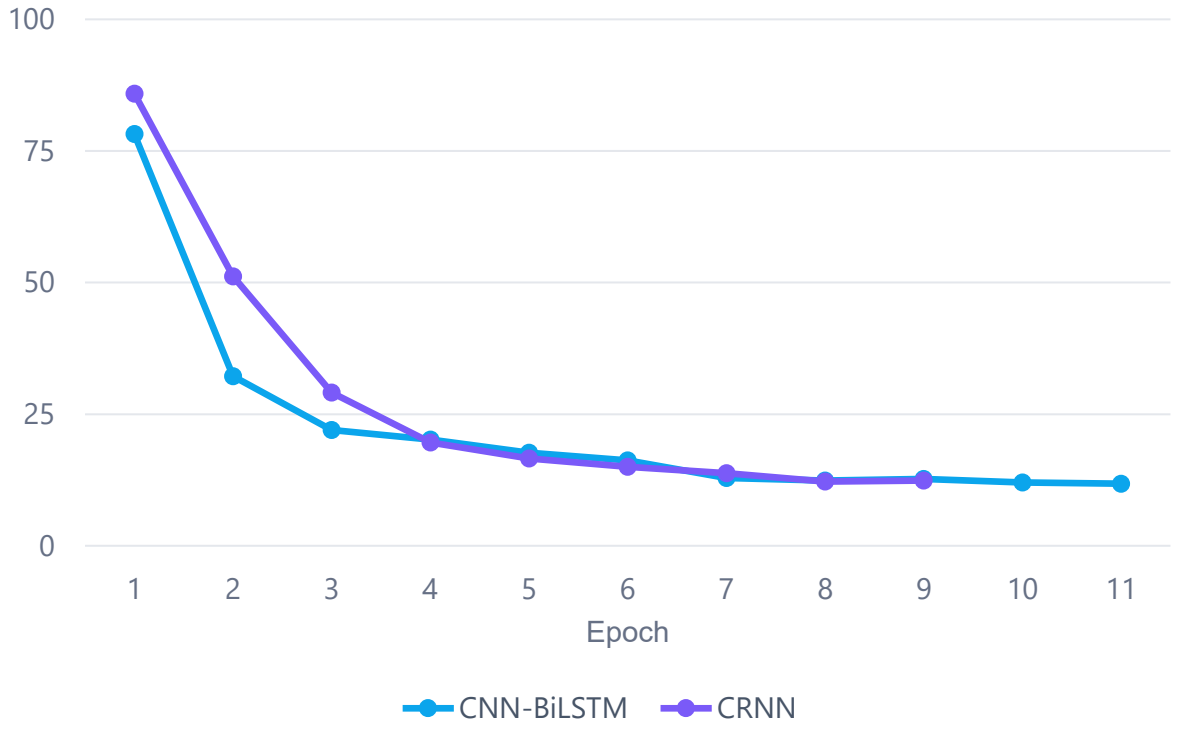
Training and fine-tuning on a laptop CPU

CNN-BiLSTM · best val CER 11.8%

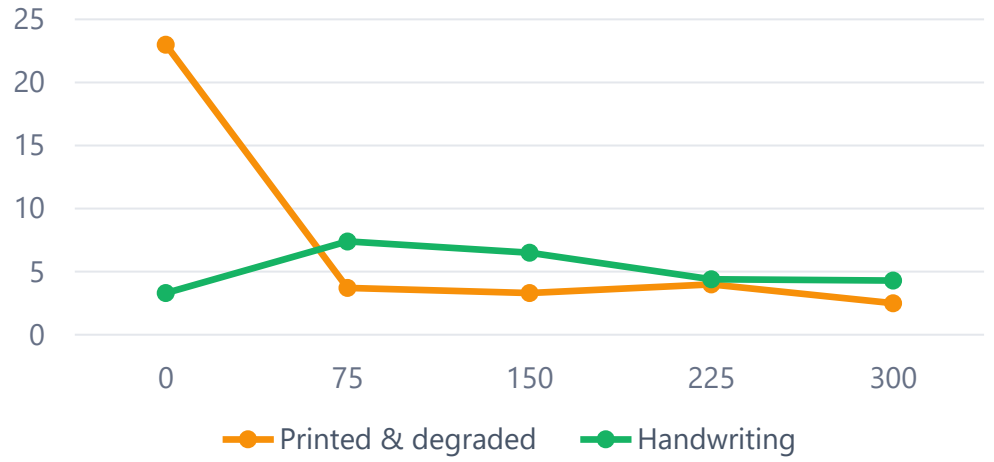
CRNN · best val CER 12.2%

TrOCR fine-tuning · 300 steps · 44 min

Validation CER (%) per epoch



TrOCR validation CER (%) by step



20.37% → 2.59%

printed-text test CER, before → after fine-tuning
handwriting unchanged: 6.97% → 6.73%

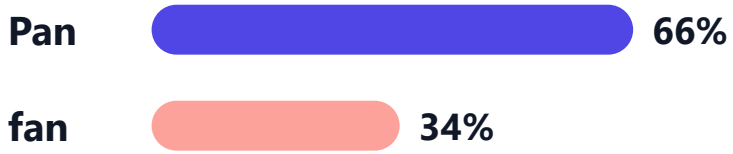
Key idea: let five models vote on every word

1. Every model reads each text line
2. The readings are aligned word by word
3. Each word is decided by a weighted vote: model weight × confidence

Words without a majority are flagged so the reviewer checks only the uncertain ones

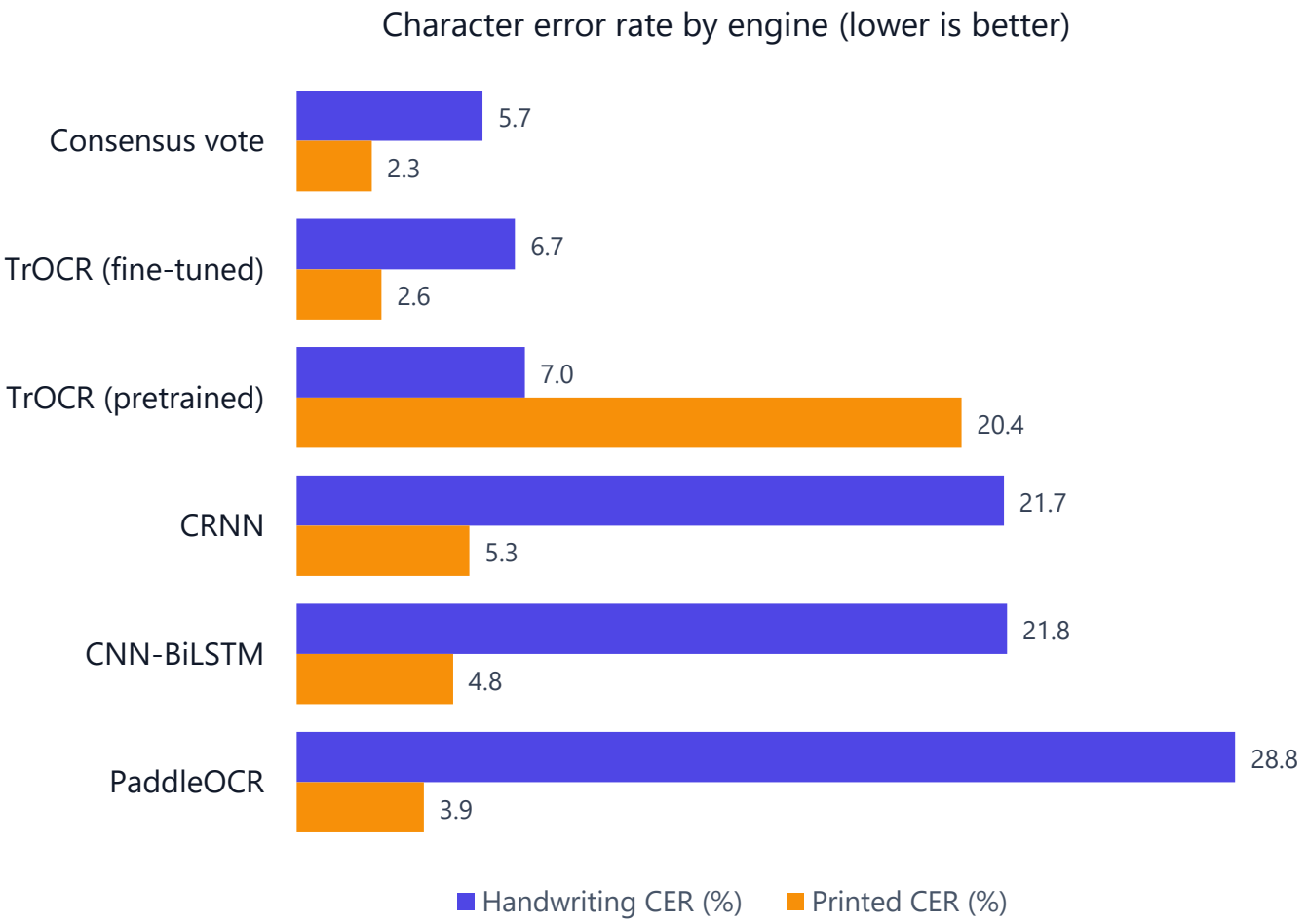
Real example from the test page: “Tin ___ Alley”

TrOCR (fine-tuned) (weight 3)	**fan**
CRNN (weight 1)	**fan**
TrOCR (pretrained) (weight 3)	**Pan**
PaddleOCR (weight 2)	**Pan**
CNN-BiLSTM (weight 2)	**Pan**



Result: “Tin Pan Alley”, which is correct

Results on 600 held-out lines per test set



5.74%

CER on handwriting from unseen writers (Consensus)

2.35%

CER on printed and degraded text (Consensus)

12 ms

per line with CNN-BiLSTM, our fastest model

Voting removes errors even the best model makes

5.74%

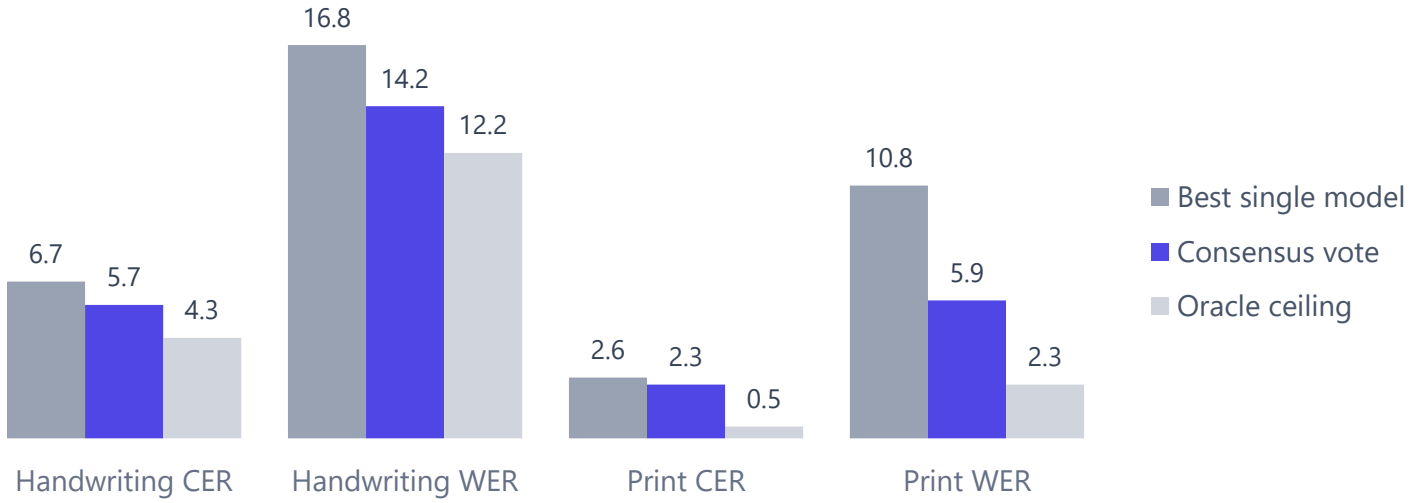
handwriting CER: 15% fewer errors than the best single model (6.73%)

5.86%

printed-text WER: 46% fewer word errors than the best single model (10.81%)

11 of 125

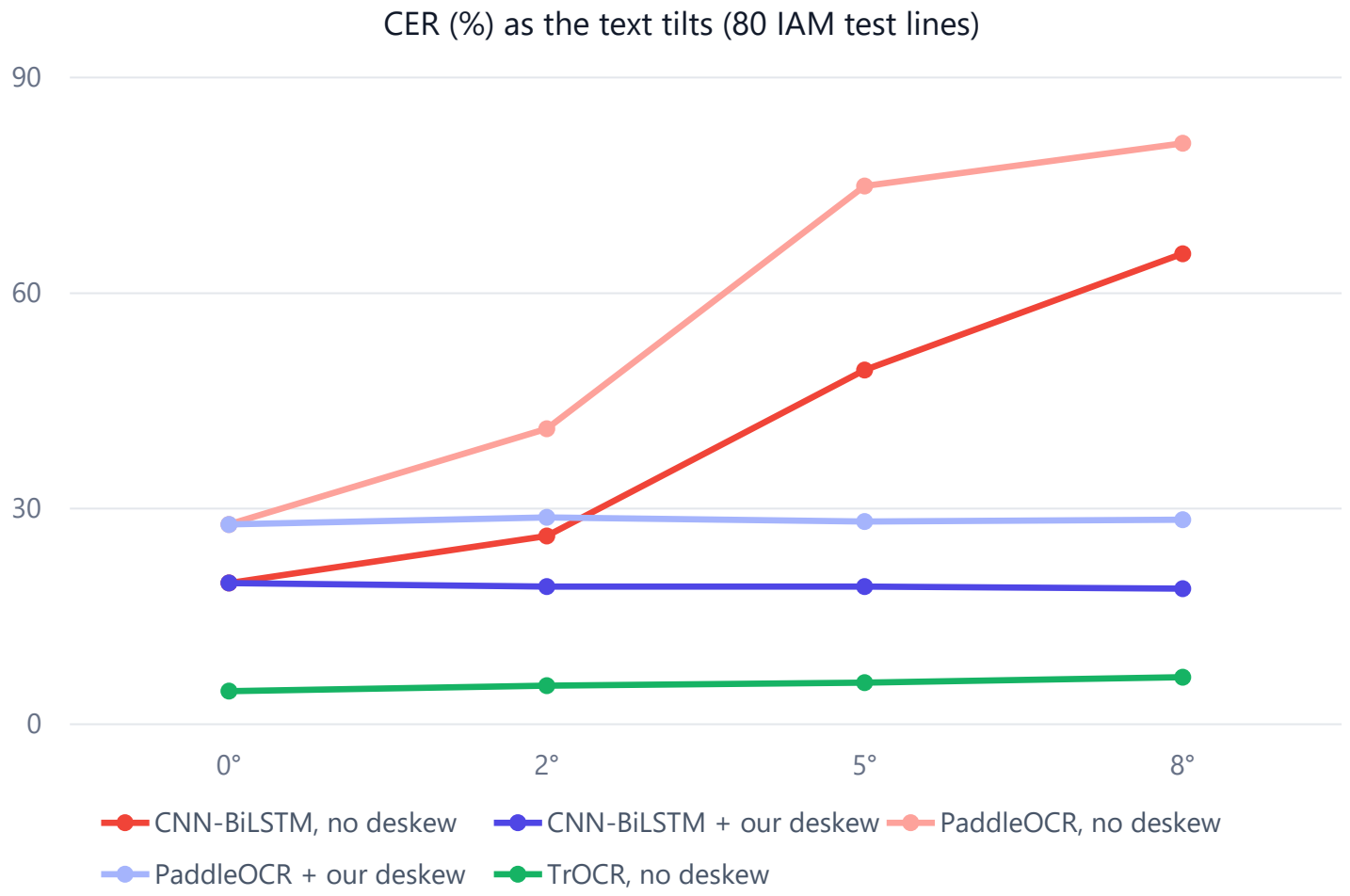
words flagged as disputed on the handwritten demo page, where most errors are



Weights tuned by cross-validation
TrOCR-ft 3 · TrOCR 3 · PaddleOCR 2 · CNN-BiLSTM 2 · CRNN 1

Cost
About 1 s per line, because every model runs

Robust to noise, and deskewing fixes rotation



Tilt breaks CTC models

CNN-BiLSTM: 19.6% → 65.5% CER at 8°

Deskew restores accuracy

18.9% at 8° with our preprocessing

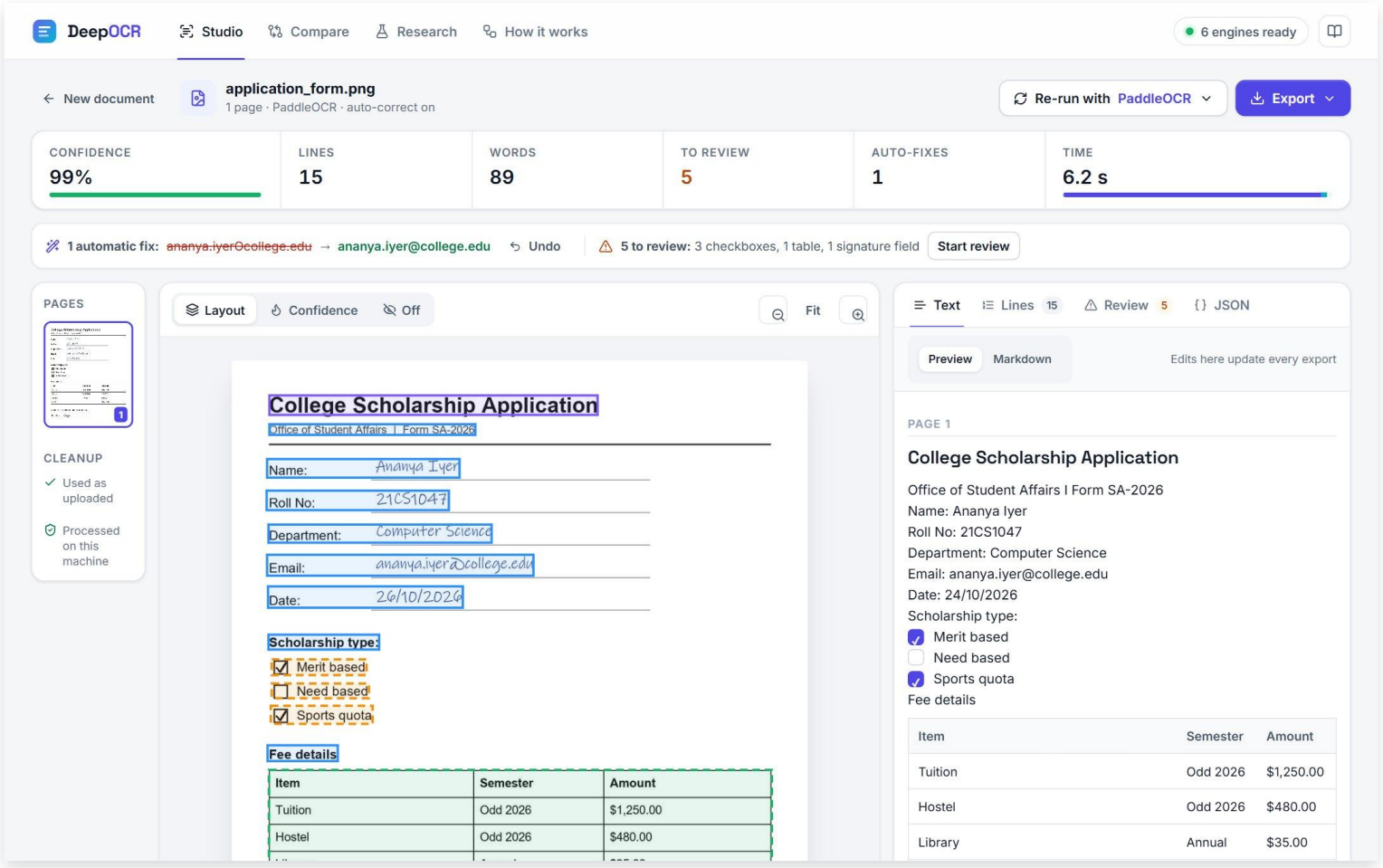
Noise barely matters

Heavy noise, 15% contrast or JPEG quality 8 change CER by under 0.2 points

TrOCR is most tilt-robust

6.5% at 8°, but most blur-sensitive

Layout analysis: from page image to structure



- **Titles and headings**
  found from line height and stroke width
- **Text and key–value rows**
  kept in reading order, column by column
- **Ruled tables**
  rebuilt as Markdown tables
- **Checkboxes**
  checked, empty or unsure
- **Signature fields**
  from a printed label (not verification)
- **Low-confidence lines**
  sent to the review queue

The DeepOCR studio: review instead of retyping

Live progress

Every stage and every model, with cancel

Disputed words

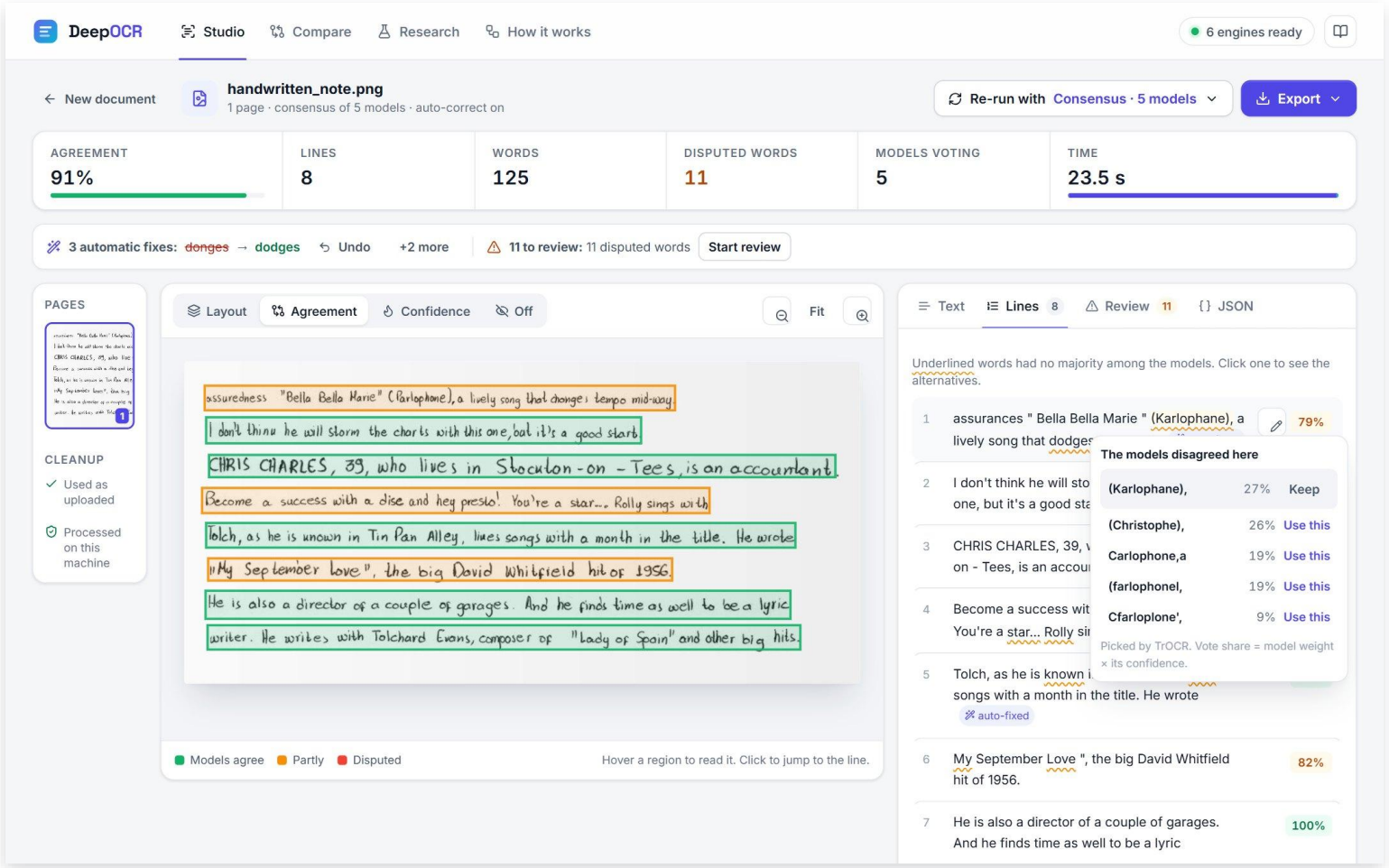
One click swaps in another model's reading

Review queue

Unclear lines, tables, checkboxes, signatures

Export

Markdown, plain text or JSON with coordinates



Also: Compare engines side by side, the Research results page, and a phone layout

Conclusion and future work

All eight proposal objectives met

- Two CTC models trained from scratch on a CPU; TrOCR fine-tuned (print CER 20.4% → 2.6%)
- Consensus vote: best accuracy (5.74% / 2.35% CER) and a guide for human review
- Complete pipeline and web studio, covered by 135 automated tests

Future work

- Hindi & multilingual
- Learned layout (LayoutLM)
- Faster, parallel voting
- Mobile camera OCR



Thank you

Questions?